Geometric Digital Twin Verification Tool

Model G(0) | Model G(t) | Verification Results

Model G(0)

Step 1. Geometric Digital Twin Characteristics

Scale

Building

Scale Value (optional)

n/d

Building Life Cycle Stage

Use B3-B5

Average Ground Resolution (AGR) of Model [mm/px]

20.764

Step 2. Sample for Verification

Sample Type

- ○ Feature-based
- ● Scale-based

Sample Scale (optional)

n/d

Step 3. Parameters for Updating (optional)

Number of Previous Updates

0

Update Schedule

Planned

Survey Type

Image-based

Data Acquisition Sequence

Mixed

Step 4. LoD Calculation

Suggested LoD: LoD 3.3 - LoD 4

Feature's RMSE (optional)

RMSE of the actual dimensions of specific elements (e.g., roof elements, windows) between the model and real-world measurements

Value

0.15

Step 5. Ground Sample Distance (GSD) Calculation

Sensor size [mm]

13.20

Focal length [mm]

10.39

Flight height [m]

80

Image width [px]

5471

Calculated GSD: 18.5773 mm

Resolution Achieved: 89.47%

Step 6. Data Quality Elements

Scale-based Evaluation

📋 Data Quality Checklist

- [x] **Positional absolute**
  Category: Mandatory | Type: Consistency | Sub-Type: Accuracy
  Alignment of the model with real-world context

🔢 Enter value for `Positional absolute`

1.250

📏 Unit for `Positional absolute`

px

- [x] **Positional relative**
  Category: Mandatory | Type: Consistency | Sub-Type: Accuracy
  Internal consistency of the model

🔢 Enter value for `Positional relative`

0.670

📏 Unit for `Positional relative`

px

- [ ] **Interoperability compliance**
  Category: Conditional | Type: Interoperability | Sub-Type: N/A
  Compliance with interoperability requirements

- [x] **LoD compliance**
  Category: Conditional | Type: Generalization | Sub-Type: N/A
  The degree to which the model meets the required LoD

Select for `LoD compliance`

- (●) Yes
- ( ) No

- [ ] **Temporal accuracy**
  Category: Optional | Type: Consistency | Sub-Type: Temporal Quality
  Accuracy of the temporal attributes of the data

Selected DQ Elements Summary

	Measure	Value	Unit	Description
0	Positional absolute	1.250	px	N/A
1	Positional relative	0.670	px	N/A
2	LoD compliance	Yes	N/A	N/A

📥 Download Model Data

💾 Save & Continue to Model G(t)

Model G(0) data saved! Please click on the 'Model G(t)' tab above to continue.

Geometric Digital Twin Verification Tool

Model G(0) | Model G(t) | Verification Results

Model G(t)

Step 1. Geometric Digital Twin Characteristics

Scale

Building

Scale Value (optional)

n/d

Building Life Cycle Stage

Use B1-B2

Average Ground Resolution (AGR) of Model [mm/px]

23.242

Step 2. Sample for Verification

Sample Type

- ○ Feature-based
- ● Scale-based

Sample Scale (optional)

n/d

Step 4. LoD Calculation

Suggested LoD: LoD 3.3 - LoD 4

Feature's RMSE (optional)

RMSE of the actual dimensions of specific elements (e.g., roof elements, windows) between the model and real-world measurements

Value

0.21

Step 5. Ground Sample Distance (GSD) Calculation

Sensor size [mm]

6.40

Focal length [mm]

4.72

Flight height [m]

80

Image width [px]

8000

Calculated GSD: 13.5593 mm

Resolution Achieved: 58.34%

Step 6. Data Quality Elements

Scale-based Evaluation

📋 Data Quality Checklist

Positional absolute

Category: Mandatory | Type: Consistency | Sub-Type: Accuracy

Alignment of the model with real-world context

🔢 Enter value for `Positional absolute`

3.140

📏 Unit for `Positional absolute`

px

Positional relative

Category: Mandatory | Type: Consistency | Sub-Type: Accuracy

Internal consistency of the model

🔢 Enter value for `Positional relative`

1.490

📏 Unit for `Positional relative`

px

LoD compliance

Category: Conditional | Type: Generalization | Sub-Type: N/A

The degree to which the model meets the required LoD

Select for `LoD compliance`

- (•) Yes
- ( ) No

Selected DQ Elements Summary

	Measure	Value	Unit	Description
0	Positional absolute	3.140	px	N/A
1	Positional relative	1.490	px	N/A
2	LoD compliance	Yes	N/A	N/A

📥 Download Model Data

💾 Save & View Results

Model G(t) data saved! Please click on the 'Verification Results' tab above to view results.

Geometric Digital Twin Verification Tool

Model G(0) | Model G(t) | Verification Results

Verification Results

Scale: Building

Building Life Cycle Stage

Model G(0): Use B3-B5

Model G(t): Use B1-B2

Sample for Verification

Selected Type: Scale-based

Sample scale: n/d

LoD Calculation Summary

Model version	AGR of model	Suggested LoD	Resolution achieved	Feature's RMSE
Model G(0)	20.764 mm/px	LoD 3.3 - LoD 4	89.47%	0.15
Model G(t)	23.242 mm/px	LoD 3.3 - LoD 4	58.34%	0.21

Data Quality Elements Summary

Measure	Model G(0)	Model G(t)
Positional absolute	1.250	3.140
Positional relative	0.670	1.490
LoD compliance	Yes	Yes

Decision Model

Condition 1: Model Accuracy Verification

Comparison of positional accuracy between Model G(0) and Model G(t)

Accuracy difference between models Dacc: 0.288

δD (Accuracy difference threshold)

0.075

Dacc vs. δD comparison: Decline in model G(t) accuracy

Models Alignment Data (optional)

Mean of geometric deviations μ

0.039

Standard deviation σ

0.146

Deviation threshold δpos (identifies significant deviations): 0.331

Condition 2: Model LoD Verification

Consistency of LoD between Model G(0) and Model G(t)

LoD Verification: Consistent LoD for both models

Condition 3: Model Resolution Verification

Consistency of resolution between Model G(0) and Model G(t)

Resolution Verification: Decline in model G(t) resolution

Performance Comparison: Model G(t) vs Model G(0) (optional)

Percentage of evaluation parameters where Model G(t) outperforms Model G(0) based on Conditional and Optional DQ Elements

Percentage of parameters where G(t) outperforms G(0): 0.0%

Interpretation: Limited improvement in Model G(t)

Verification Score

⚠️ Warning

Only one of three conditions is fulfilled. Decline in metrics for model G(t). Unsuitable for the updating process.

Verification Legend

- ❌ **Critical**: Model G(t) shows significant decline in all key metrics. Attention required.
- ⚠️ **Warning**: Model G(t) shows decline in multiple metrics. Review and improvement needed.
- ⚠️ **Partial**: Model G(t) shows mixed results. Consider partial updates with specific improvements.
- ✅ **Suitable**: Model G(t) shows overall improvement. Ready for full update implementation.

Download Verification Results

📥 Download Results (CSV) | 📥 Download Results (JSON)